3 **(a)** State two differences between stationary waves and progressive waves.

1. .....

.....

2. .....

..... [2]

(b) **(i)** A laser produces a narrow beam of coherent light of wavelength 632 nm. The beam is incident normally on a diffraction grating as shown in Fig. 3.1.

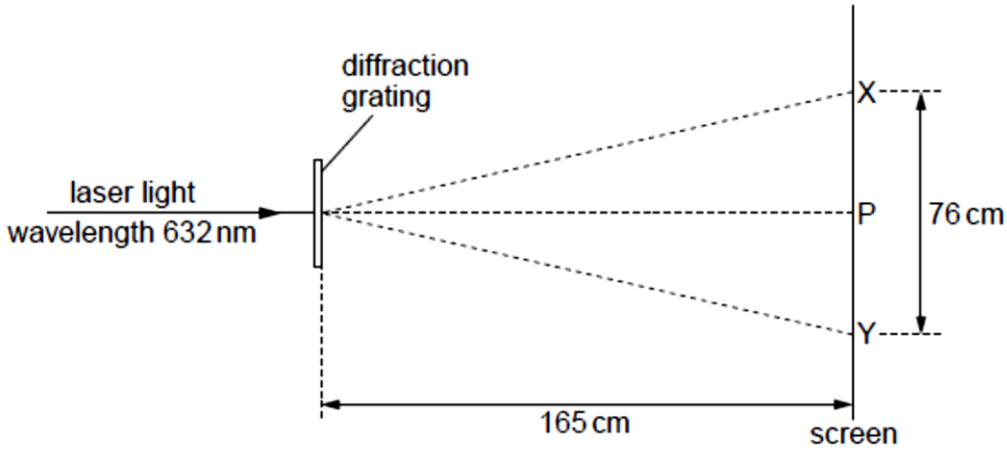

Fig. 3.1 (Top view)

Spots of light are observed on a screen placed parallel to the grating. The distance between the grating and the screen is 165 cm.

The brightest spot is P. The spots formed closest to P and on each side of P are X and Y.

X and Y are separated by a distance of 76 cm.

Calculate the number of lines per metre on the grating.

number per metre = ... m^{-1} [2]

(ii) The grating in **(b)(i)** is now rotated about an axis parallel to the incident laser beam, as shown in Fig. 3.2.

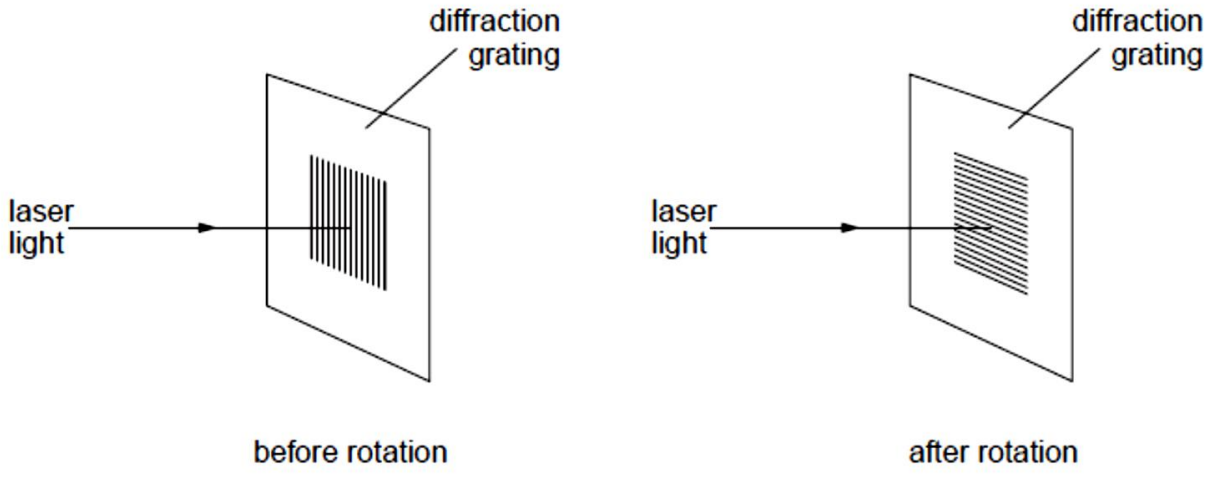

Fig. 3.2

State what effect, if any, this rotation will have on the positions of the spots P, X and Y.

.....

..... [2]

(iii) In another experiment using the apparatus in **(b)(i)**, a student notices that the distances XP and PY, as shown in Fig. 3.1 are not equal.

Suggest a reason for this difference.

.....

.....

..... [1]

(c) A cord is held under tension between two fixed points A and B, as shown in Fig. 3.3. The distance AB is 0.40 m.

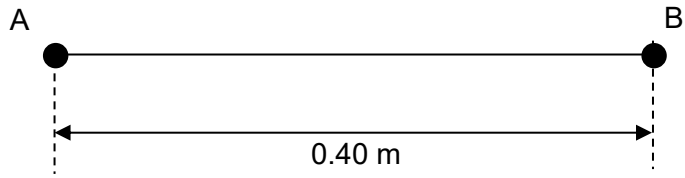

Fig. 3.3

(i) Explain why only stationary waves of certain frequencies are able to form between A and B.

.....

.....

..... [1]

(ii) The string is made to resonate in a mode with the third lowest possible frequency. Calculate the wavelength of this wave.

wavelength = ... m [1]

(iii) By reference to the formation of the stationary wave, explain the significance of the product of frequency and wavelength for a stationary wave.

.....

.....

.....

..... [2]